9b.Simulation of Space Wave Propagation Characteristics using MATLAB

Overall MATLAB Code:

```
clc;

clear;

close all;

%% Objective 1 : Space Wave Propagation

f = 900; % Frequency (MHz)

Pt = 30; % Transmit Power (dBm)

d = 1:1:100; % Distance (km)

FSPL = 32.44 + 20*log10(f) + 20*log10(d);

Pr = Pt - FSPL;

%% Objective 2 : LOS and Radio Horizon

ht = 50; % Transmitter Height (m)

hr = 25; % Receiver Height (m)

d_LOS = 4.12*(sqrt(ht)+sqrt(hr));

%% Objective 3 : Link Budget Analysis

Gt = 10; % Tx Gain (dB)

Gr = 8; % Rx Gain (dB)

Pr_link = Pt + Gt + Gr - FSPL;

%% Display Results

fprintf('\nSPACE WAVE PROPAGATION ANALYSIS\n');

fprintf('---------------------------------\n');

fprintf('Operating Frequency = %.0f MHz\n',f);


fprintf('LOS Distance = %.2f km\n',d_LOS);
```

```
fprintf('Received Power at 10 km = %.2f dBm\n',Pr(10));
%% Plot Results
figure('Name',...
'Space Wave Propagation Characteristics',...
'NumberTitle','off');
subplot(2,2,1)
plot(d,FSPL,'LineWidth',2)
grid on
xlabel('Distance (km)')
ylabel('FSPL (dB)')
title('Free Space Path Loss')
subplot(2,2,2)
plot(d,Pr,'LineWidth',2)
grid on
xlabel('Distance (km)')
ylabel('Received Power (dBm)')
title('Received Power vs Distance')
subplot(2,2,3)
bar(d_LOS)
grid on
title('LOS Communication Range')
ylabel('Distance (km)')
set(gca,'XTickLabel',{'LOS Range'})
subplot(2,2,4)
plot(d,Pr_link,'LineWidth',2)
grid on
```

```
xlabel('Distance (km)')

ylabel('Power (dBm)')

title('Link Budget Analysis')
```

output: